

MI-26: Resolution

Matthew J. Colbrook

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge, United Kingdom
m.colbrook@damtp.cam.ac.uk

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope.](#)

1 MI-26: concavity alone does not give two-unitary subadditivity

The target asks whether, for every real-valued concave function f on $[0, \infty)$ with $f(0) \geq 0$ and every positive semidefinite A, B , there are unitaries U, V such that

$$f(A + B) \leq Uf(A)U^* + Vf(B)V^*. \quad (1)$$

It is explicitly the removal of monotonicity that is at issue [1, Problem 5]; see also [2, Remark 3.13].

Theorem 1.1. *The assertion (1) is false in dimension two. A counterexample is provided by the polynomial $f(t) = t - t^2$ and two real rank-one projections. It remains false when both inputs are required to be positive definite.*

Proof. The function $f(t) = t - t^2$ is concave on $[0, \infty)$ and satisfies $f(0) = 0$. Let

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \frac{1}{25} \begin{pmatrix} 9 & 12 \\ 12 & 16 \end{pmatrix}.$$

Both matrices are orthogonal projections, so $f(P) = f(Q) = 0$. However,

$$f(P + Q) = P + Q - (P + Q)^2 = -(PQ + QP) = \frac{1}{25} \begin{pmatrix} -18 & -12 \\ -12 & 0 \end{pmatrix}.$$

Its eigenvalues are $6/25$ and $-24/25$; in particular $(1, -2)^T$ is an eigenvector for $6/25$. Thus $f(P + Q) \not\leq 0$, whereas the right-hand side of (1) is zero for every U, V .

For positive definite inputs, take

$$A = P + \frac{1}{20}I_2, \quad B = Q + \frac{1}{20}I_2.$$

Each has eigenvalues $21/20$ and $1/20$, and consequently

$$\lambda_{\max}(f(A)) = \lambda_{\max}(f(B)) = \frac{19}{400}.$$

The eigenvalues of $P + Q$ are $8/5$ and $2/5$. Thus $A + B$ has eigenvalues $17/10$ and $1/2$, giving

$$\lambda_{\max}(f(A + B)) = \frac{1}{4}.$$

For every pair of unitaries,

$$Uf(A)U^* + Vf(B)V^* \leq \frac{19}{200}I_2.$$

Since $1/4 > 19/200$, domination (1) is again impossible. $\square$

Remark 1.2 (The function class matters). This counterexample uses a concave function that is real-valued, not nonnegative on the whole half-line. It does not contradict the theorem for nonnegative concave functions. In fact, a concave function that is nonnegative everywhere on $[0, \infty)$ must be nondecreasing: a negative secant slope, by concavity, would force the function eventually to become negative. The repository target and the cited open question allow real-valued concave functions, so $f(t) = t - t^2$ belongs to the stated class.

References

- [1] Koenraad M. R. Audenaert and Fuad Kittaneh. Problems and conjectures in matrix and operator inequalities. arXiv:1201.5232, 2012. Version consulted: v3; Problem 5, printed page 3, and Problem 7, printed pages 12–13. URL: <https://arxiv.org/abs/1201.5232>.
- [2] Jean-Christophe Bourin and Eun-Young Lee. Unitary orbits of Hermitian operators with convex or concave functions. arXiv:1109.2384, 2011. Remark 3.13. URL: <https://arxiv.org/abs/1109.2384>.